Research Note 7

1 what is $\int \frac{d\sigma}{d\Omega} d\Omega p(1-cos\theta)$

$\sigma = P * S$,$d\sigma = dP * S$, which means the probability(dimensionless) of an event to happen in the area where $d\sigma$ represents.
So $dP * N * p(1-cos\theta)$ is the total momentum transfer inside $d\sigma$, and the total momentum transfer is $\int \frac{d\sigma}{S} Np(1-cos\theta)\frac{d\Omega}{d\Omega} = \sigma_T * p * N/S$

2 where does $sin^2\theta$ come from?

$$\psi \to \psi_{ij}$$

Moment $M_\psi(t) \equiv \int d^3v\, \psi(\vec{v}) f(\vec{v},t) = \int d^3v\, \psi_{ij}(\vec{v}) f(\vec{v})$

$$\partial_t f = C[f] = \int d^3v_*\, d\Omega\, g\sigma(f'f'_* - ff_*)$$

$$\int \partial_t f \cdot \psi_{ij}\, d^3v = \int d^3v\, d^3v_*\, d\Omega\, g\sigma(f'f'_* - ff_*)\psi_{ij}$$

$$\partial_t \int d^3v\, \psi_{ij} f = \int \dots$$

$$\partial_t M_{\psi_{ij}} = \int d^3v\, \psi_{ij} C[f]$$

Linearization of $C[f]$

$$f = f^{(0)} + \delta f = f^{(0)}(1+\Phi), \quad \Phi \equiv \frac{\delta f}{f^{(0)}}$$

Then:

$$\begin{aligned} f'f'_* - ff_* &= f^{(0)\prime}(1+\Phi')f^{(0)\prime}_*(1+\Phi'_*) - f^{(0)}(1+\Phi)f^{(0)}_*(1+\Phi_*) \\ &= f^{(0)\prime}f^{(0)\prime}_*(1+\Phi'+\Phi'_*) - f^{(0)}f^{(0)}_*(1+\Phi+\Phi_*) \end{aligned}$$

Detailed balance gives:

$$f^{(0)}f^{(0)}_* = f^{(0)\prime}f^{(0)\prime}_*$$

$$f^{(0)} = n\left(\frac{m\beta}{2\pi}\right)^{\frac{3}{2}} e^{-\beta\varepsilon}$$

$$f^{(0)}f^{(0)}_* = n^2\left(\frac{m\beta}{2\pi}\right)^3 e^{-\beta(\varepsilon+\varepsilon_*)} = \dots e^{-\beta\left[\frac{1}{2}m(v^2+v_*^2)\right]}$$

But $v^2 + v_*^2 = v'^2 + v'^2_*$ (Elastic collision / Conservation of energy and momentum)

$$\begin{aligned} \to &= f^{(0)}f^{(0)}_*(\Phi' + \Phi'_* - \Phi - \Phi_*) \\ &\equiv f^{(0)}f^{(0)}_*\Delta\Phi \end{aligned}$$

$$\therefore C_L[f] = \int d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)} \Delta\Phi$$

$$\text{And } \partial_t M_{\psi_{ij}} = \int d^3v\, \psi_{ij} C_L[f]$$

$$= \int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)} \Delta\Phi \psi_{ij}(\vec{v})$$

Collision $\leftrightarrow$ Inverse collision: $(\vec{v}, \vec{v}_*) \leftrightarrow (\vec{v}', \vec{v}_*')$

$$\rightarrow \int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)}(\vec{v}') f_*^{(0)}(\vec{v}_*') \Delta\Phi_u \psi_{ij}(\vec{v}')$$

$$\Delta\Phi = \Phi(\vec{v}') + \Phi(\vec{v}_*') - \Phi(\vec{v}) - \Phi(\vec{v}_*)$$

$$\Delta\Phi_u = \Phi(\vec{u}) + \Phi(\vec{u}_*) - \Phi(\vec{u}') - \Phi(\vec{u}_*')$$

$$\rightarrow = \int \underbrace{d^3u\, d^3u_*}_{J=1}\, d\Omega\, g\sigma\, \underbrace{f^{(0)}(\vec{u}) f_*^{(0)}(\vec{u}_*)}_{\text{Detailed Balance}} \Delta\Phi_u \psi_{ij}(\vec{u}')$$

Dummy variable transformation $u \rightarrow v$:

$$\rightarrow = \int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)}(\vec{v}) f_*^{(0)}(\vec{v}_*)(-\Delta\Phi)\psi_{ij}(\vec{v}')$$

$$\therefore 2\dot{M}_{\psi_{ij}} = \int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)} \Delta\Phi(\psi_{ij}(\vec{v}) - \psi_{ij}(\vec{v}'))$$

By $\vec{v} \leftrightarrow \vec{v}_*$:

$$2\dot{M}_{\psi_{ij}} = \int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)} \Delta\Phi(\psi_{ij}(\vec{v}_*) - \psi_{ij}(\vec{v}_*'))$$

$$\therefore 4\dot{M}_{\psi_{ij}} = \int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)} \Delta\Phi(-\Delta\psi_{ij})$$

To show what is Φ,

$$\frac{\partial f^{(0)}}{\partial t} + \vec{v} \cdot \vec{\nabla} f^{(0)} = C_L[f^{(0)}\Phi]$$

$$f = f^{(0)} + f^{(1)} + \cdots = -\beta f^{(0)} \psi_{ij} \frac{\partial u_i}{\partial x_j} \cong f^{(0)}(1 + \Phi)$$

Where $\frac{\partial u_i}{\partial x_j}$ is the second-order velocity gradient tensor.

$$f^{(0)} = n \left(\frac{m\beta}{2\pi}\right)^{\frac{3}{2}} e^{-\frac{1}{2}m(\vec{v}-\vec{u})^2\beta}$$

$$\partial_t f^{(0)} = \frac{\partial f^{(0)}}{\partial n}\frac{\partial n}{\partial t} + \frac{\partial f^{(0)}}{\partial \vec{u}} \cdot \frac{\partial \vec{u}}{\partial t} + \frac{\partial f^{(0)}}{\partial T}\frac{\partial T}{\partial t}$$

$$\vec{\nabla} f^{(0)} = \frac{\partial f^{(0)}}{\partial n}\vec{\nabla} n + \frac{\partial f^{(0)}}{\partial \vec{u}} \cdot \vec{\nabla}\vec{u} + \frac{\partial f^{(0)}}{\partial T}\vec{\nabla} T$$

Using the Euler equations (0th-order approximation):

$$\begin{cases} \frac{\partial n}{\partial t} + \vec{\nabla} \cdot (n\vec{u}) = 0 & \text{(Continuity)} \\ nm\frac{\partial \vec{u}}{\partial t} + nm(\vec{u} \cdot \vec{\nabla})\vec{u} = -\vec{\nabla}(nkT) & \text{(Momentum)} \\ \frac{3}{2}nk\frac{\partial T}{\partial t} + \frac{3}{2}nk(\vec{u} \cdot \vec{\nabla})T = -nkT\vec{\nabla} \cdot \vec{u} & \text{(Energy / Adiabatic)} \end{cases}$$

$$\vdots$$

$$\frac{\partial f^{(0)}}{\partial t} + \vec{v}\cdot\vec{\nabla} f^{(0)} = -\beta f^{(0)}\psi_{ij}\frac{\partial u_i}{\partial x_j} + (\text{trace / unrelated terms, not related to shear})$$

$$\psi_{ij} = m\left(C_i C_j - \frac{1}{3}\delta_{ij}C^2\right)$$

Here $\vec{C} = \vec{v} - \vec{u}$ is the peculiar velocity (and $\vec{C}_* = \vec{v}_* - \vec{u}$), which will cancel out or integrate out in $\Delta\psi_{ij}$.

$$\frac{\partial u_i}{\partial x_j} = \frac{1}{3}\delta_{ij}\vec{\nabla}\cdot\vec{u} + \text{antisymmetric part} + D_{ij} \quad \text{(Traceless symmetric)}$$

Since $\psi_{ij}\cdot\text{trace} = \sum_i \psi_{ii} = 0$ and $\psi_{ij}\cdot\text{antisymmetric} \equiv 0$, only $\psi_{ij}D_{ij}$ remains.

$$\Phi(\vec{v}) = \sum_{l=0}^{\infty}\sum_{m=-l}^{l}\varphi_{lm}(|\vec{v}|)Y_{lm}(\hat{v})$$

$$\begin{aligned} C_L[\Phi] &= \int d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)}\Delta\Phi \\ &= \int d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)}\sum\sum\varphi_{lm}Y_{lm} + \int\dots \end{aligned}$$

Because it is independent of direction, only the same l mode contributes to this rotationally invariant space. No specific coordinate axes are fixed, so physical laws remain invariant. If $l \neq l'$, orthogonality forces it to 0.

Hence, $C_L[\Phi] = C_L[\varphi Y_{lm}] = f\cdot[\varphi Y_{lm}]$, 1 is invariant)

Since:$C_L[f^{(0)}\Phi] = -\beta f^{(0)}\psi_{ij}D_{ij}$, the right hand side is a scalar but with $l = 2$ behavior,

And C_L is a linear operator, $f^{(0)}\Phi$ must strictly belong to the $l = 2$ mode. Otherwise, other modes would appear on the right-hand side.$\psi \implies l = 2 \implies \therefore \Phi = \lambda\psi_{ij}$

Infact it is Sonine polynomials:

$$\Phi = \psi_{kl}\sum_{n=0}^{\infty}a_n S_n(v^2) \cong \psi_{kl}\lambda \quad \text{(Taking the } n = 0 \text{ term)}$$

$$\Delta\Phi = \Phi' + \Phi'_* - \Phi - \Phi_* = \lambda(\psi'_{kl} + \psi'_{kl*} - \psi_{kl} - \psi_{kl*}) = \lambda\Delta\psi_{kl}$$

$$\therefore \dot{M}_{\psi_{ij}}[\lambda\psi_{kl}] = -\frac{1}{4}\lambda\int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)}\Delta\psi_{kl}\Delta\psi_{ij}$$

$$\lambda\dot{M}_{\psi_{ij}}[\psi_{kl}] = -\frac{1}{4}\lambda\int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)}\Delta\psi_{kl}\Delta\psi_{ij}$$

Independent of λ. Material properties should not depend on λ, which is why λ needs to be separated out.

$\dot{M}_{\psi_{ij}}[\psi_{kl}]$ is a fourth-rank tensor. Consider when the first two indices and the last two indices are identical. In fact, it becomes $\Delta\psi_{ij}\cdot\Delta\psi_{ij}$, which is analogous to a vector inner product. This represents the change in a specific direction, or the projection of $\Delta\psi_{kl}$ onto ψ_{ij}. We do not care about a component changing into other directional components. Then, summing over indices eliminates coordinate dependence, making it rotationally invariant (representing the total change amplitude across all directions).

$$\text{Therefore, } \sum_{ij}\dot{M}_{\psi_{ij}}[\psi_{ij}] \equiv -\frac{1}{4}\int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)}\sum_{ij}(\Delta\psi_{ij})^2$$

$$\text{Scalar } I = -\frac{1}{4}\int d^3v\, d^3v_*\, d\Omega\, g\sigma f^{(0)} f_*^{(0)}\sum_{ij}(\Delta\psi_{ij})^2$$

$$\sum_{ij}(\Delta\psi_{ij})^2 = \frac{m^2}{4}(g_i'g_j' - g_ig_j)^2$$

$$= \frac{m^2}{4}\sum_{ij}\left[(g_i'g_j')^2 - 2g_i'g_j'g_ig_j + (g_ig_j)^2\right]$$

Let:

$$\vec{g} = (0, 0, g)$$
$$\vec{g}' = (g\sin\theta\cos\varphi, g\sin\theta\sin\varphi, g\cos\theta)$$

When $i = 1$:

$$\frac{m^2}{4}\left[(g\sin\theta\cos\varphi \cdot g\sin\theta\cos\varphi)^2 + (g\sin\theta\cos\varphi \cdot g\sin\theta\sin\varphi)^2 + (g\sin\theta\cos\varphi \cdot g\cos\theta)^2\right]$$

When $i = 2$:

$$\frac{m^2}{4}\left[(g\sin\theta\sin\varphi \cdot g\sin\theta\cos\varphi)^2 + (g\sin\theta\sin\varphi \cdot g\sin\theta\sin\varphi)^2 + (g\sin\theta\sin\varphi \cdot g\cos\theta)^2\right]$$

When $i = 3$:

$$\frac{m^2}{4}\left[(g\cos\theta \cdot g\sin\theta\cos\varphi)^2 + (g\cos\theta \cdot g\sin\theta\sin\varphi)^2 + (g\cos\theta \cdot g\cos\theta)^2 - 2g\cos\theta g\cos\theta g^2 + g^4\right]$$

Expanding all terms:

$$\begin{aligned}\text{Expansion} = \frac{m^2}{4}g^4\Big[&\sin^4\theta\cos^4\varphi + \sin^4\theta\cos^2\varphi\sin^2\varphi + \sin^2\theta\cos^2\varphi\cos^2\theta \\ &+ \sin^4\theta\sin^2\varphi\cos^2\varphi + \sin^4\theta\sin^4\varphi + \sin^2\theta\sin^2\varphi\cos^2\theta \\ &+ \cos^2\theta\sin^2\theta\cos^2\varphi + \cos^2\theta\sin^2\theta\sin^2\varphi + \cos^4\theta - 2\cos^2\theta + 1\Big]\end{aligned}$$

Since $(\cos^2\theta - 1)^2 = \sin^4\theta$:

$$\begin{aligned}&= \frac{m^2g^4}{4}\left(\sin^4\theta\cos^2\varphi + \sin^4\theta\sin^2\varphi + \sin^2\theta\cos^2\theta + \cos^2\theta\sin^2\theta + \sin^4\theta\right) \\ &= \frac{m^2g^4}{4}\left(\sin^4\theta + 2\sin^2\theta\cos^2\theta + \sin^4\theta\right) \\ &= \frac{m^2g^4}{4}\cdot 2\sin^2\theta(\sin^2\theta + \cos^2\theta) \\ &= \frac{m^2g^4}{2}\sin^2\theta\end{aligned}$$